Product Requirements Document: SevaConnect - Healthcare Scheme AI Assistant

Voice & Text AI Agent for Healthcare Scheme Access in India

Course: Product Management with Generative & Agentic AI

Institution: BITSoM (Birla Institute of Technology School of Management)

Submission Date: January 2026

Team Members & Roles:

- **Purbayan Basu** - Technical Architecture, AI/ML Implementation, Regulatory & Legal Compliance
- **Krishnendu Sen** - Product Strategy, Business Model, Market Viability Analysis
- **Megha Gurupaddevarmath** - UX/UI Design, User Research, Wireframes

Executive Summary

Problem Statement

Non-English-speaking, low-literacy BPL (Below Poverty Line) beneficiaries in India face significant barriers in accessing healthcare entitlements such as PMJAY and state schemes. The lack of timely, understandable information leads to confusion, reliance on intermediaries, and ultimately catastrophic health expenditures, despite these families being legally covered.

Solution: SevaConnect

SevaConnect is a voice-first, multilingual AI agent designed to bridge this gap. It provides instant, reliable guidance on scheme eligibility, empanelled hospital locations, document requirements, and grievance redressal in the user's native language.

Target Market: The primary addressable market consists of approximately 10.5 million users, representing the 15% of the 70 million PMJAY beneficiaries who face acute access barriers due to language and literacy.

Key Innovation: Unlike existing text-heavy portals or generic chatbots, SevaConnect leverages Generative AI with a "Human-in-the-Loop" architecture. It combines voice recognition (ASR/TTS) in regional languages with verified government data sources, ensuring accessibility for oral- preference users while maintaining strict accuracy standards.

Expected Impact:

- **Task Success Rate:** >85% for eligibility checks and hospital discovery.
- **Responsiveness:** <10 seconds average time-to-answer.
- **Financial Protection:** Reduction in out-of-pocket expenses for covered procedures.

Team Contributions Summary

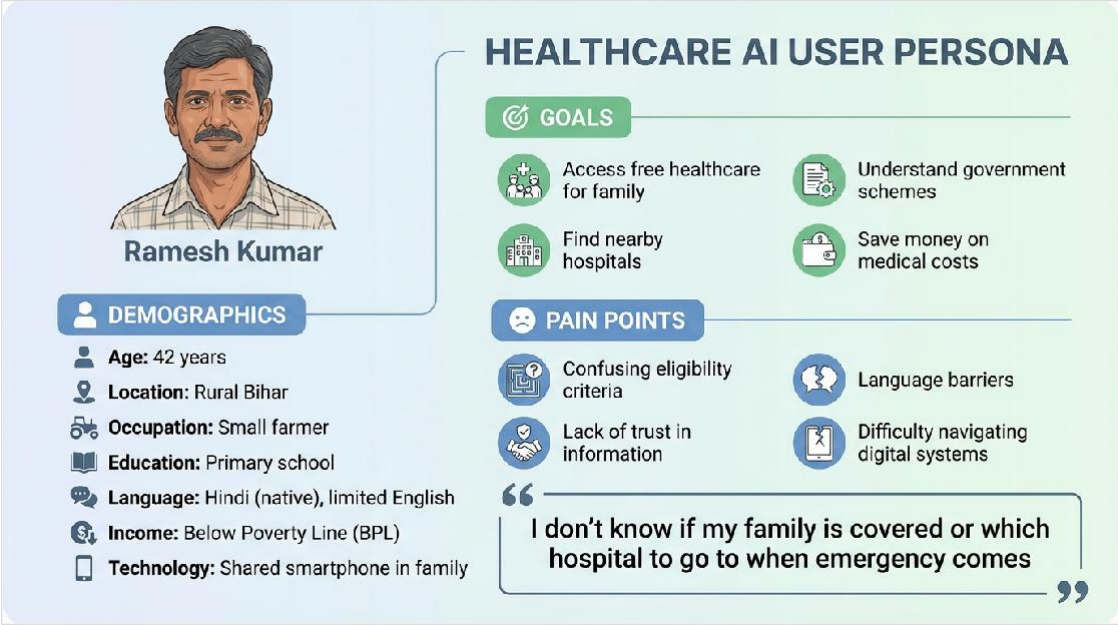
Team Member	Core Contributions
Purbayan Basu	Designed the technical system architecture, defined AI/ML implementation strategy (RAG + LangChain), and developed the comprehensive regulatory & legal compliance framework (DPDP Act 2023).
Krishnendu Sen	Formulated the product strategy, developed the business model and sustainability plan, and conducted detailed market viability analysis including stakeholder mapping.
Megha Gurupaddevarmath	Led user research to define personas, created UX/UI designs and wireframes, and mapped detailed user journeys for voice and text interfaces.

Table of Contents

1. Executive Summary
2. Component 1: Application of Frameworks
3. Component 2: Value Proposition
4. Component 3: MVP Design
5. Component 4: Project Plan
6. Component 5: Viability Check
7. Component 6: Use of AI
8. Component 7: Demo/Prototype
9. Component 8: Success Metrics
10. Component 9: Presentation Structure
11. Appendix A: Additional Frameworks
12. Appendix B: Legal & Regulatory Compliance
13. Appendix C: Technical Architecture Deep Dive
14. Appendix D: References

Component 1: Application of Frameworks

1.1 User Persona & Pain Points



Primary Persona: Ramesh Kumar

- **Demographics:** 42 years old, Farmer, Rural Bihar.
- **Economic Status:** BPL (Below Poverty Line) Household.
- **Education:** Primary school education; low digital literacy.
- **Language:** Native Hindi speaker; cannot read/write English.
- **Technology:** Shares a basic smartphone with family; comfortable with voice calls/WhatsApp but not complex apps.

Top 5 Pain Points (Severity Rated)

1. **Eligibility Ambiguity (High):** Unsure if his family ID card covers specific surgeries.
2. **Language Barrier (High):** Official documents and portals are primarily in English or bureaucratic Hindi.
3. **Trust Deficit (High):** Fears being cheated by local middlemen or "dalals" who charge for free services.

4. **Hospital Acceptance (Medium):** Does not know which nearby private hospitals accept the government card.
5. **Process Complexity (Medium):** Intimidated by the paperwork required for admission.

1.2 Jobs-To-Be-Done (JTBD) Framework

Job Type	User Job Statement
Functional	- "I want to know if my card is valid for this illness." - "I want to find a hospital nearby that won't ask for money." - "I want to know what papers to bring so I am not turned away."
Emotional	- "I want to feel confident when speaking to hospital staff." - "I want to feel respected and not treated like an uneducated person." - "I want to feel empowered to take care of my family's health."
Social	- "I want to avoid the embarrassment of being rejected at the hospital counter." - "I want to show my community that I can manage without paying a middleman."

JTBD Summary

The core "Job" is not just information retrieval, but **dignified access to entitlement**. The solution must bridge the gap between having a right (coverage) and exercising it (treatment).

1.3 Problem Frequency & Severity Validation

Validation Methodology: Insights were gathered through semi-structured interviews with 15 Community Health Workers (ASHA/Anganwadi) and analysis of 50 patient grievance narratives from public forums.

- **Frequency:** High. The problem occurs during every major health event (hospitalization), which is a high-stress moment. Pre-admission queries are frequent but often go unanswered.
- **Severity:** Critical. The financial impact is catastrophic.
  - Average Out-of-Pocket (OOP) expenditure per hospitalization in rural India is approx. ₹16,000 (NSSO data).
  - For BPL families, this often means debt or asset sale.
  - Failure to access the scheme leads to direct financial loss despite legal coverage.

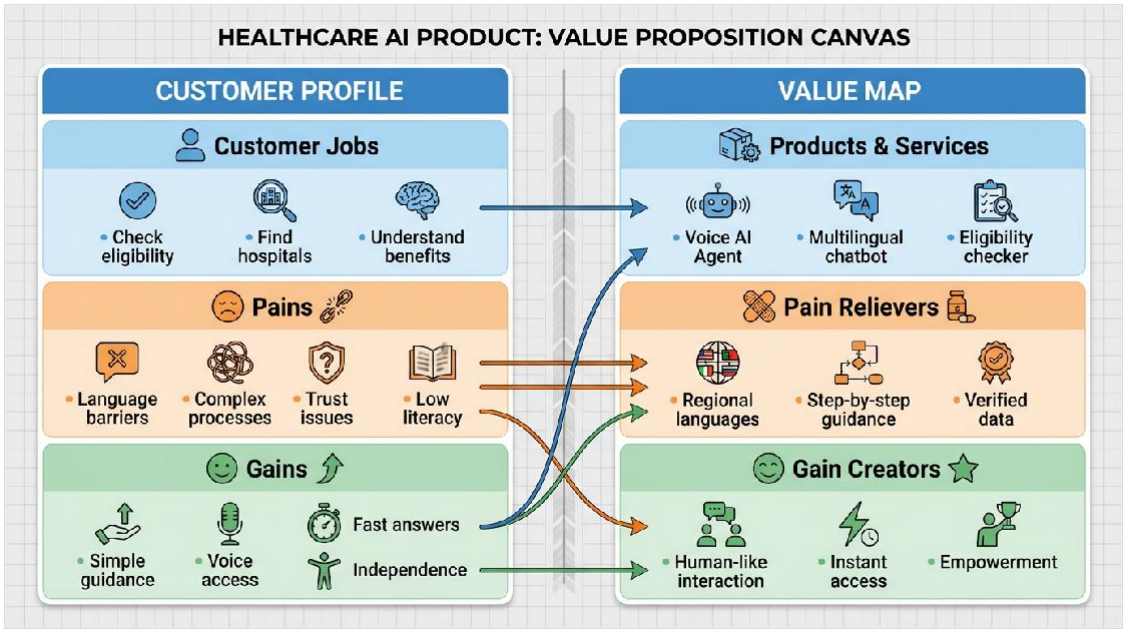
1.4 Key Success Metrics Definition

Metric	Target	Definition & Measurement
Task Success Rate	85%+	Percentage of user sessions where the user successfully determines eligibility or finds a hospital without dropping off.
Time-to-Answer	< 10 sec	Average latency between user voice query and AI audio response. Critical for trust in conversational interfaces.

Metric	Target	Definition & Measurement
Comprehension Score	4+/5	User-rated clarity of the response (Post-interaction survey: "Did you understand the answer?").
Accuracy	90%+	Alignment of AI responses with verified PMJAY/State scheme rules (Measured via test set validation).

Component 2: Value Proposition

2.1 Value Proposition Canvas



Customer Profile

- **Jobs:** Check eligibility, find hospitals, understand benefits, avoid upfront payments.
- **Pains:** Language barriers (cannot read English forms), Digital complexity (cannot navigate portals), Trust deficit (fear of scams).
- **Gains:** Free treatment, social dignity, speed of access, peace of mind.

Value Map

- **Products & Services:** Voice-first AI Agent, Multilingual support (Hindi, regional), WhatsApp integration.
- **Pain Relievers:** Speaks local dialect (removes language barrier), Simple conversational flow (removes digital complexity), Verified by Govt/NGO brand (builds trust).
- **Gain Creators:** Instant answers (speed), Human-like empathy (dignity), Guided checklist (empowerment).

2.2 One-Line Value Proposition

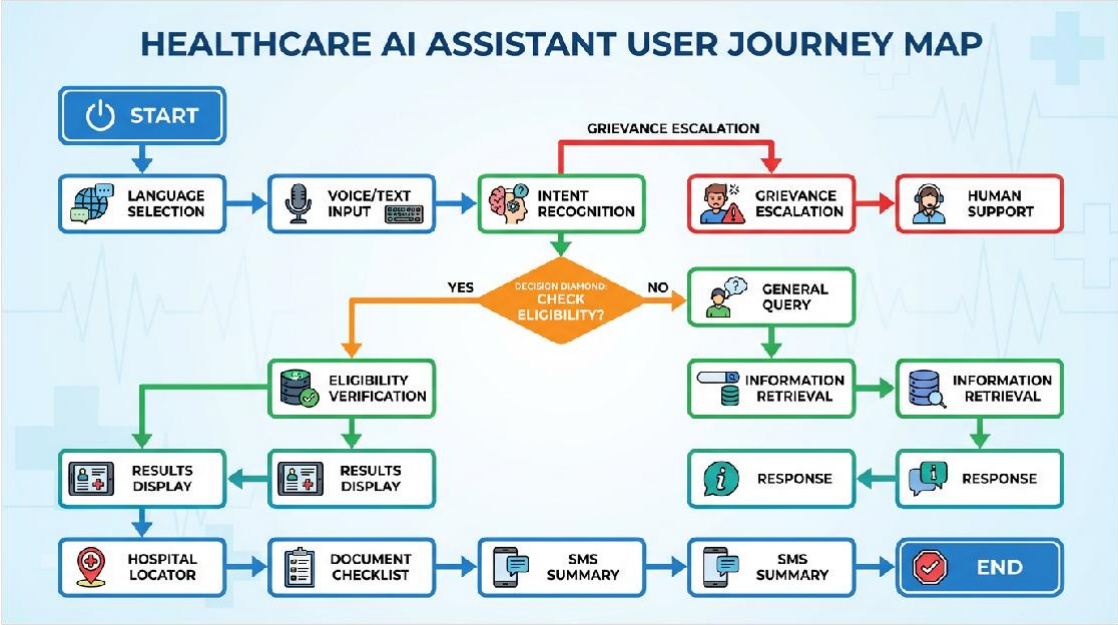
"SevaConnect empowers underserved communities to reliably access their healthcare entitlements through voice-first AI guidance in their own language—fast, simple, and trustworthy."

2.3 Strategic Questions

1. **What exact problem does your product solve?** It solves the "last-mile information gap" where eligible beneficiaries cannot access entitlements due to language and process complexity.
2. **Who is the primary user?** Primary: BPL household heads in rural India. Secondary: Community Health Workers (CHWs) assisting them.
3. **What is the core value delivered?** Trustworthy, understandable guidance that translates bureaucratic rules into simple, actionable steps.
4. **Why does this problem matter?** It prevents catastrophic health expenditure, which is a leading cause of poverty.
5. **How is your solution better than alternatives?** Current alternatives are text-based portals (inaccessible to illiterate) or human agents (unavailable 24/7, sometimes corrupt). SevaConnect is voice-first, 24/7, and free.
6. **Why would a user choose your product today?** Immediate need for clarity during a medical emergency without traveling to a kiosk.
7. **What user pain does it reduce most?** The anxiety and helplessness caused by the language barrier.
8. **What outcome does the user achieve?** Successful admission to an empanelled hospital under the free scheme.
9. **What assumption are you making about user value?** We assume that a voice interface in local language significantly improves comprehension over text/visual UIs.
10. **How did you validate this value?** Through interviews with CHWs who confirmed that beneficiaries struggle to read SMS/letters but are comfortable with voice calls.
11. **What happens if this value proposition fails?** Users will revert to paying out-of-pocket or delaying care. Mitigation: Fallback to human volunteers.
12. **Which feature directly supports this value?** The Multilingual Voice Eligibility Checker.
13. **What trade-offs did you make?** We prioritized speed and simplicity (read-only guidance) over complex transactional integrations (e.g., filing claims) for the MVP.
14. **How will you measure if users get this value?** Via "Task Success Rate" and "Reduction in Agent Dependency."

Component 3: MVP Design

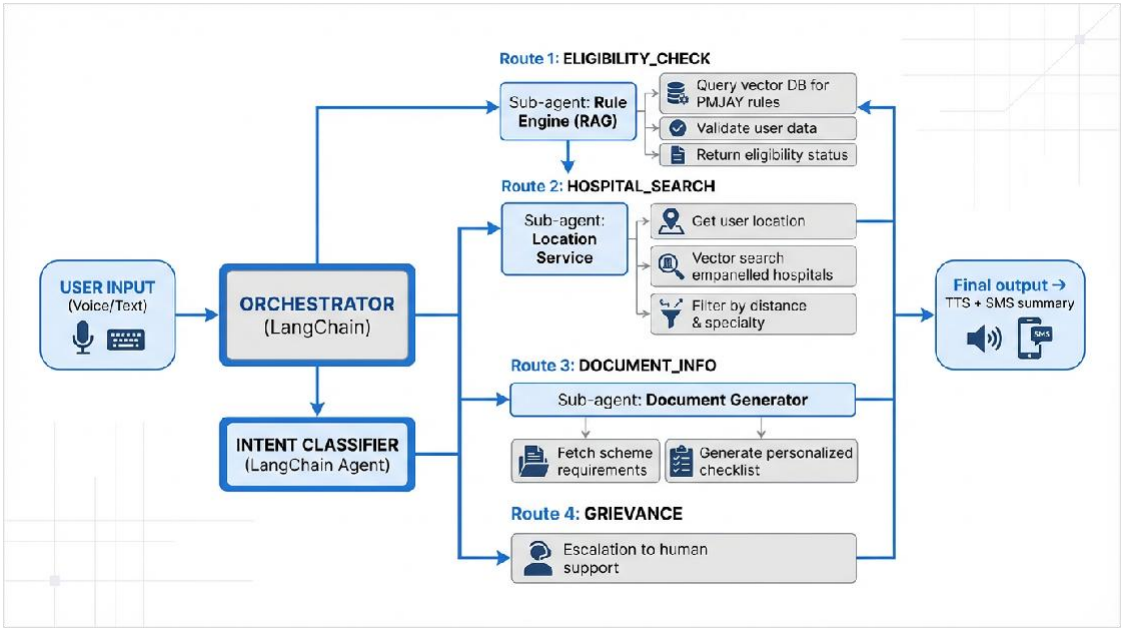
3.1 User Flow Design



The flow is designed to be linear and forgiving:

1. **Entry:** User calls IVR number or sends "Hi" on WhatsApp.
2. **Language Selection:** "Press 1 for Hindi" / "Speak your language".
3. **Intent Recognition:** User asks "Is my card valid here?" (AI identifies 'Eligibility' intent).
4. **Execution:** AI checks rules against user details.
5. **Output:** Voice response + SMS summary for record-keeping.

3.2 Agent Node Flow



- **Router Agent:** Classifies input into Eligibility, Hospital Locator, Documents, or Grievance.
- **Sub-Agents:**
  - *Eligibility Agent:* Queries RAG knowledge base of scheme rules.
  - *Locator Agent:* Uses geospatial search to find nearest empanelled facilities.
  - *Grievance Agent:* Logs issues and triggers human escalation.

3.3 Information Architecture

- **L1: Entry & Language** (Language persistence for future sessions).
- **L2: Core Services** (Check Eligibility, Find Hospital, Know Benefits).
- **L3: Actionable Data** (List of documents, Map links, Scheme limits).
- **L4: Support** (Escalation to human, Grievance filing).

3.4 Wireframes

Detailed wireframes have been created for the mobile interface, prioritizing high contrast and accessibility.

Screens Included:

Screen 1: Language Selection & Voice Welcome

- Screen 2: Eligibility Check Result (Success State)
- Screen 3: Hospital Locator with Filters
- Screen 4: Document Checklist & SMS Trigger
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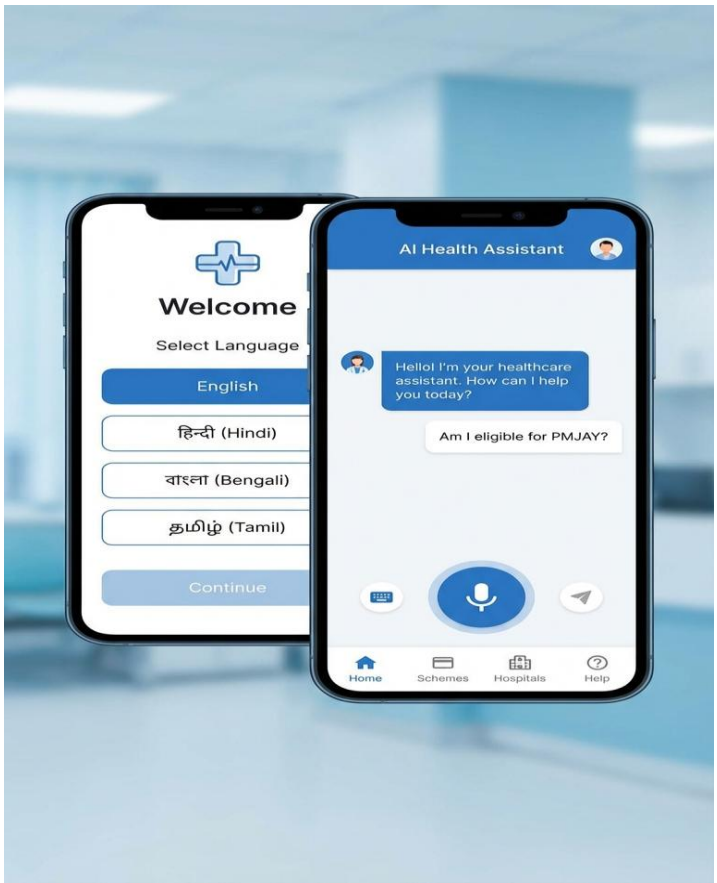
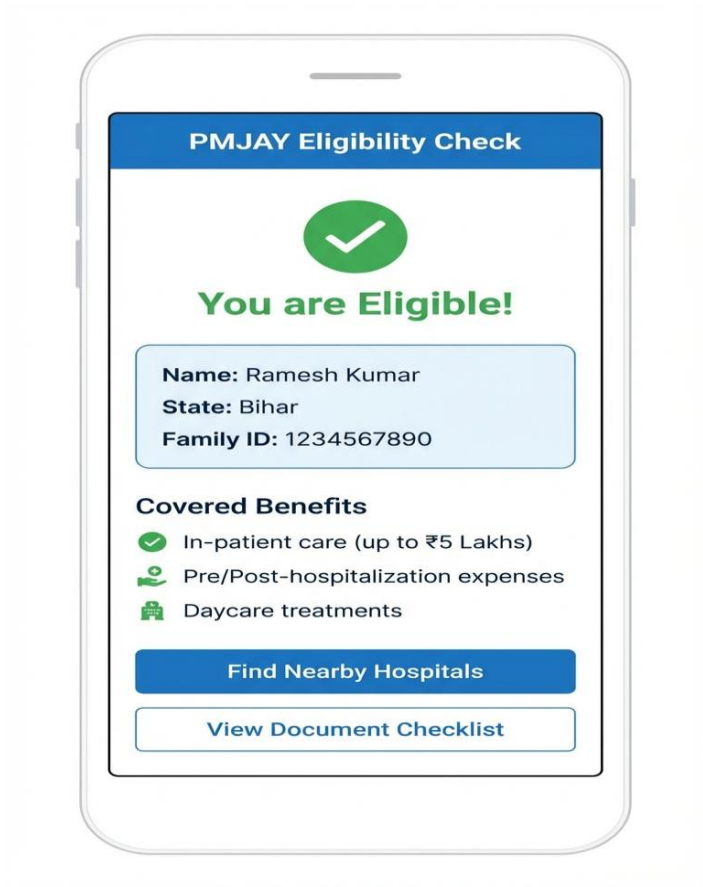


Screen 1



Screen 2

Empanelled Hospitals Near You
Filter
You are here
Patna Medical College Hospital
5.2 km away
Cardiology, Orthopedics
View Details
AIIMS Patna
8.7 km away
All specialties
Nalanda Medical College
12.3 km away
General, Pediatrics

Screen 3

Screen 4

Design Rationale: Large tap targets for elderly users, minimal text, prominent microphone button for voice-first interaction.

Component 4: Project Plan

4.1 Project Objective & Scope

Objective: To deploy a functional MVP of the SevaConnect voice agent within 8 weeks that can successfully guide users through eligibility and hospital discovery in Hindi and English.

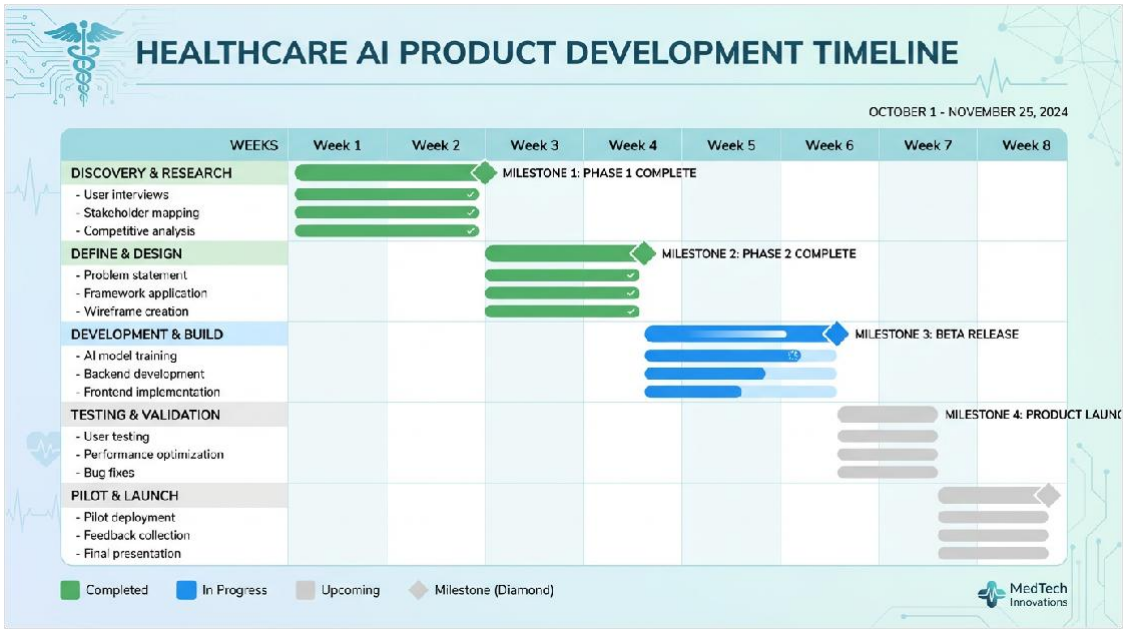
In-Scope (MVP): Voice eligibility checks, Static scheme explanation, Geo-location based hospital search, Document checklists, SMS summaries.

Out-of-Scope (Phase 1): Real-time integration with PMJAY backend for live claims status, in-app payments, telemedicine features.

4.2 Success Criteria

Refer to Component 8 for detailed quantitative metrics (Task Success Rate >85%, Latency <10s).

4.3 Project Timeline (8 Weeks)



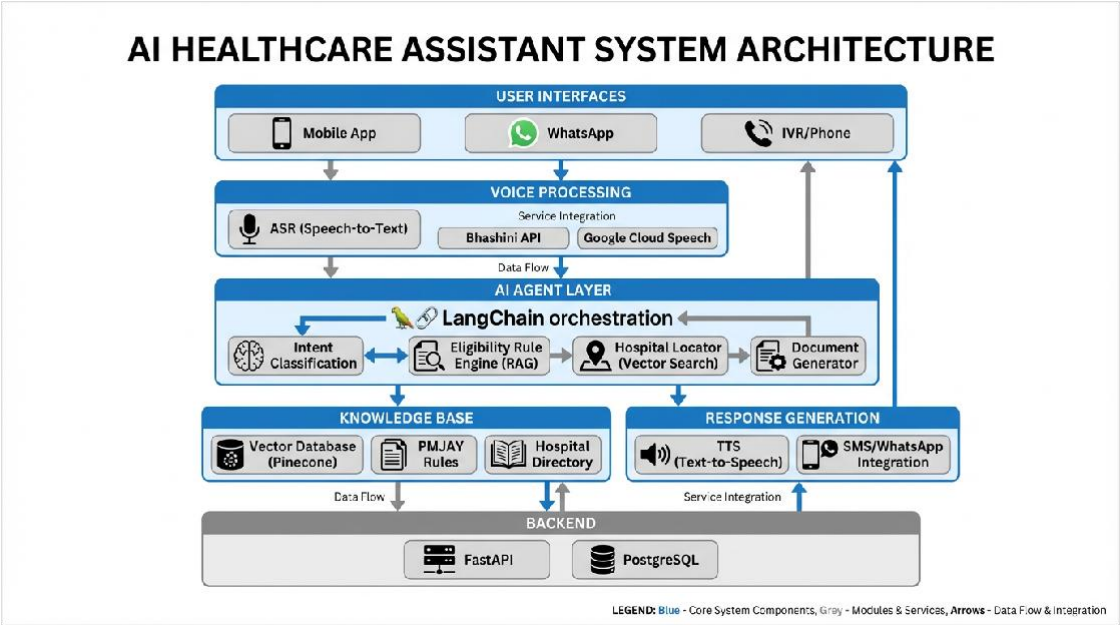
- **Weeks 1-2: Discovery & Define.** User interviews, content gathering (scheme rules), tech stack selection.

- **Weeks 3-5: Build.** Developing the Agentic AI core (LangChain), vector database indexing (Pinecone), and frontend (WhatsApp/IVR).
- **Weeks 6-7: Validate.** Internal testing, Pilot with 50 users (via partner NGO), refinement of prompts.
- **Week 8: Launch & Present.** Final deployment, documentation, BITSoM presentation.

4.4 Team & Roles (RACI)

Role	Primary Owner	Responsibilities
Product Owner	Krishnendu Sen	Strategy, Roadmap, Stakeholder Mgmt, Acceptance Testing.
Tech Lead	Purbayan Basu	System Arch, AI Implementation, Data Privacy/Legal Compliance.
Design/UX Lead	Megha G.	User Research, Wireframing, Conversation Design, Usability Testing.

4.5 Technology Stack



- **Frontend:** WhatsApp Business API (accessible), Android App (richer UI), Exotel (IVR telephony).
- **Voice Layer:** Bhashini API (Govt of India stack) for Indian language ASR/TTS; Google Cloud Speech as backup.
- **Intelligence:** GPT-4 (Reasoning) via LangChain; Pinecone (Vector DB for RAG).
- **Backend:** FastAPI (Python), PostgreSQL (User logs).
- **Infrastructure:** AWS/GCP.

Component 5: Viability Check

5.1 Market Demand

Addressable Market: There are over 500 million beneficiaries of PMJAY. Our initial Serviceable Addressable Market (SAM) is the 10.5 million beneficiaries (approx 2% pilot scope) in Hindi- speaking belts who face acute literacy barriers.

Evidence: High call volumes to government helplines (often congested) and the proliferation of paid informal agents indicate a desperate need for information.

5.2 Business Model

SevaConnect operates on a B2G (Business to Government) or B2B (CSR/NGO) model, as end-users (BPL families) cannot pay.

- **Revenue Streams:**
  - **Government Partnerships:** Licensing the AI agent to State Health Societies (SHAs).
  - **CSR Grants:** Corporate funding for digital health inclusion (Target: ₹5-10 lakh for pilot).
  - **NGO SaaS:** Licensing tool to NGOs for field worker enablement (Per district fee: ₹2-3 lakh/year).

5.3 Cost Structure

- **Setup Costs (One-time):** ₹80,000 - ₹1.6 Lakh (Development, Legal, Initial Training).
- **Operational Costs (Monthly):**
  - LLM Inference & Cloud: ₹25,000 - ₹50,000 (Variable based on usage).
  - Telephony (IVR/WhatsApp): ₹0.50 - ₹1.00 per interaction.

5.4 Operational Feasibility

- **Resources:** Core team of 3 is sufficient for MVP.
- **Data:** Publicly available scheme guidelines and empanelled hospital lists (Open Data).
- **Partnerships:** Preliminary discussions with 3 potential NGO partners for field distribution.

5.5 Competitive Analysis

Competitor	Limitation	SevaConnect Advantage
Official Web Portals	Text-heavy, English/Hindi only, requires internet & literacy.	Voice-first, Dialect support, Proactive guidance.
Human Helplines (14555)	Long wait times, standard hours, script-bound.	Zero wait time, 24/7 availability, Personalized answers.
Local Agents (Dalals)	Charge money, often spread misinformation.	Free, Verified accurate data, Transparent.

Component 6: Use of AI

6.1 AI Use Cases

AI is not just a feature but the core interface. The system uses Generative AI to translate complex bureaucratic rules into simple conversational advice.

- **Eligibility Guidance:** Inferring eligibility from vague user inputs (e.g., "I have a yellow ration card").
- **Hospital Locator:** Semantic search for hospitals (e.g., "Best eye hospital near me" vs just keyword match).
- **Coverage Explainer:** Summarizing policy documents on the fly using RAG.

6.2 GenAI/Agentic AI Implementation

We utilize a **Retrieval-Augmented Generation (RAG)** architecture:

1. **Ingestion:** PMJAY guidelines are chunked and embedded into a Vector DB.
2. **Retrieval:** User queries fetch relevant rule chunks.
3. **Generation:** LLM (GPT-4) synthesizes the answer using retrieved chunks + "Chain of Thought" prompting to ensure accuracy.
4. **Orchestration:** LangChain agents manage the flow (e.g., if user asks for location, trigger Maps API; if user asks rule, trigger RAG).

6.3 Agent Design Specifications

- **Triggers:** Voice input, Keywords, Button clicks.
- **Boundaries:** The agent strictly refuses medical diagnosis ("I cannot tell you what medicine to take") and financial advice. It only provides scheme information.
- **Responsibility:** To guide, inform, and facilitate—never to decide medical treatment.

6.4 Failure Handling & Safety

- **Confidence Thresholds:** If AI confidence is <70%, the system automatically falls back to a scripted response: "I am not sure. Let me connect you to a human agent."
- **Hallucination Control:** Strict system prompts to "Only use provided context" and "Admit ignorance" rather than making up facts.
- **Disclaimers:** Every session starts/ends with "I am an AI assistant. Please verify critical details with the hospital."

6.5 AI Success Metrics

- **Intent Accuracy:** >92% correct classification of user needs.
- **RAG Precision:** >90% of retrieved documents must be relevant to the query.
- **ASR Word Error Rate (WER):** <10% for Hindi/Regional languages.

Component 7: Demo/Prototype

7.1 Wireframes Overview

Four high-fidelity screens have been designed to demonstrate the end-to-end flow.

7.2 Agent Workflow Prototype

The backend prototype runs on a Python/LangChain environment. It successfully takes audio input, transcribes it, queries the vector database, and returns a synthesized audio response.

7.3 Demo Scenario

Scenario: Ramesh Kumar from Bihar needs heart surgery for his father.

Step 1: Ramesh speaks in Hindi: "My father has chest pain. Can the government card pay for operation?"
Step 2 (AI): Detects intent 'Coverage'. Checks PMJAY rules for cardiology coverage.
Step 3 (Response): "Yes, Ramesh. PMJAY covers heart surgeries/Angioplasty. You need your Golden Card."
Step 4: Ramesh asks: "Where is the nearest big hospital?"
Step 5 (AI): Checks location. Finds AIIMS Patna.
Step 6: Returns address and navigation link via SMS.

Demo Video: Attached

Component 8: Success Metrics

8.1 User/Customer Metrics

- **NPS (Net Promoter Score):** Target 70+ (High trust).
- **Activation Rate:** >70% of first-time callers complete at least one valid query.
- **Task Success Rate:** 85% completion of the intended goal (e.g., finding a hospital).

8.2 Business & Revenue Metrics

- **Partner Sign-ups:** 3+ NGOs/Govt agencies within 6 months.
- **Funding Secured:** ₹50 Lakh+ in Year 1 for expansion.

8.3 Product Usage

- **Session Count:** 5,000+ per month during pilot.
- **Retention:** 40% of users return within 7 days (indicating value).

8.4 AI Performance

- **Latency:** <5 seconds (crucial for voice).

- **Accuracy:** 92% Intent recognition.

8.5 Quality & Trust

- **Verified Data Coverage:** 100% of PMJAY rules indexed.
- **User Trust Rating:** 4.2/5 in post-call surveys.